

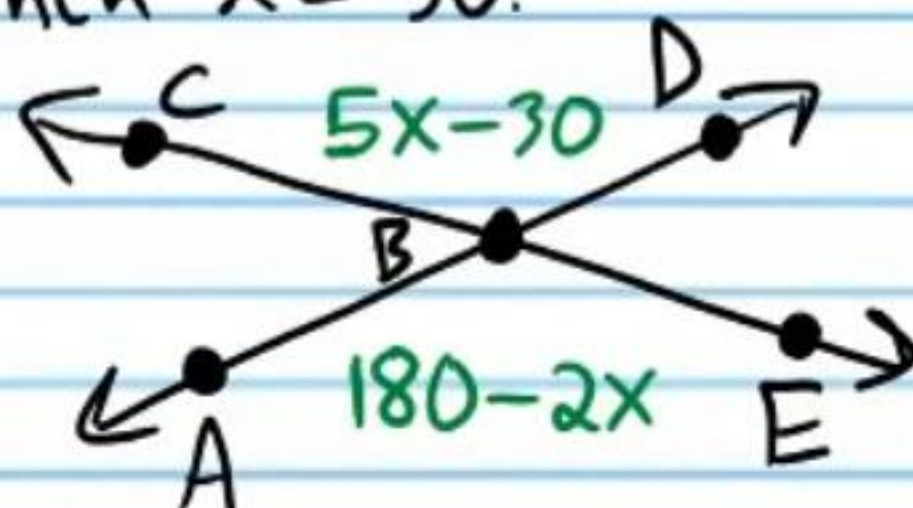
Proofs and Properties of Equality and Congruence (Homework)

Prove the following statements with two-column proofs.

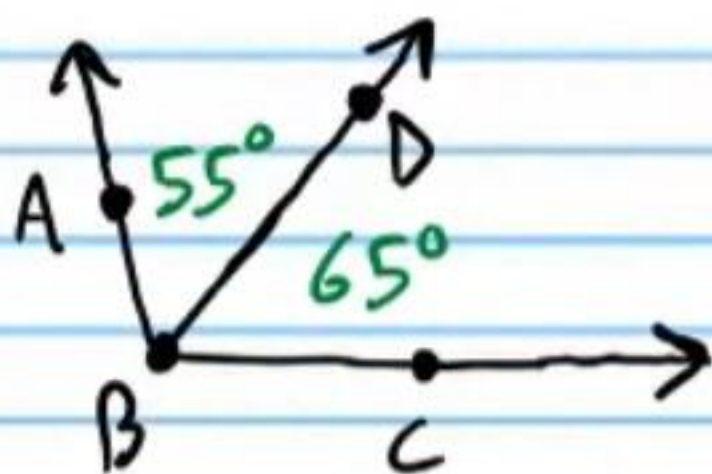
- ① If $AB = 15$ and $BC = 3$ in the segment below, then $AC = 18$.



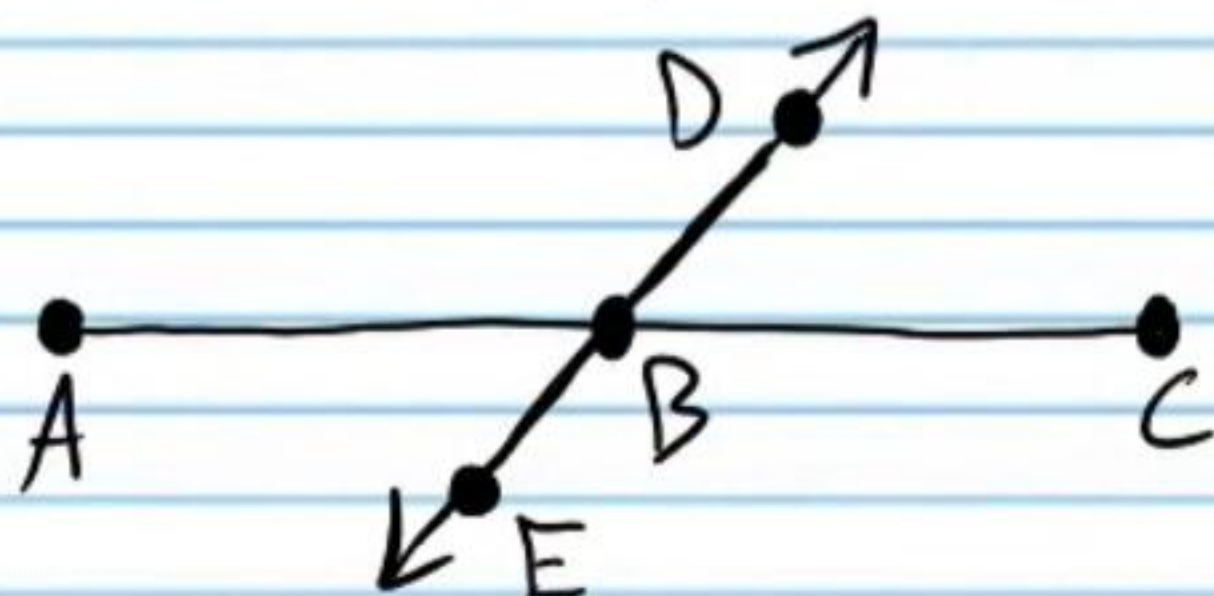
- ② If $\angle ABE$ and $\angle CBD$ are vertical angles in the figure below, then $x = 30$.



- ③ If $m\angle ABD = 55^\circ$ and $m\angle DBC = 65^\circ$, then $m\angle ABC = 120^\circ$.

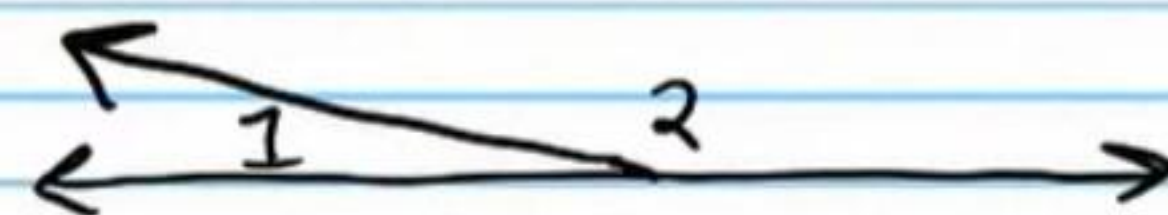


- ④ If $\overline{AB} \cong \overline{BC}$ in the diagram below, then $\overleftrightarrow{DE}$ bisects $\overline{AC}$ (this is the converse of problem #18 in class).



- ⑤ If $\overline{AB} \cong \overline{CD}$, then $\overline{CD} \cong \overline{AB}$.

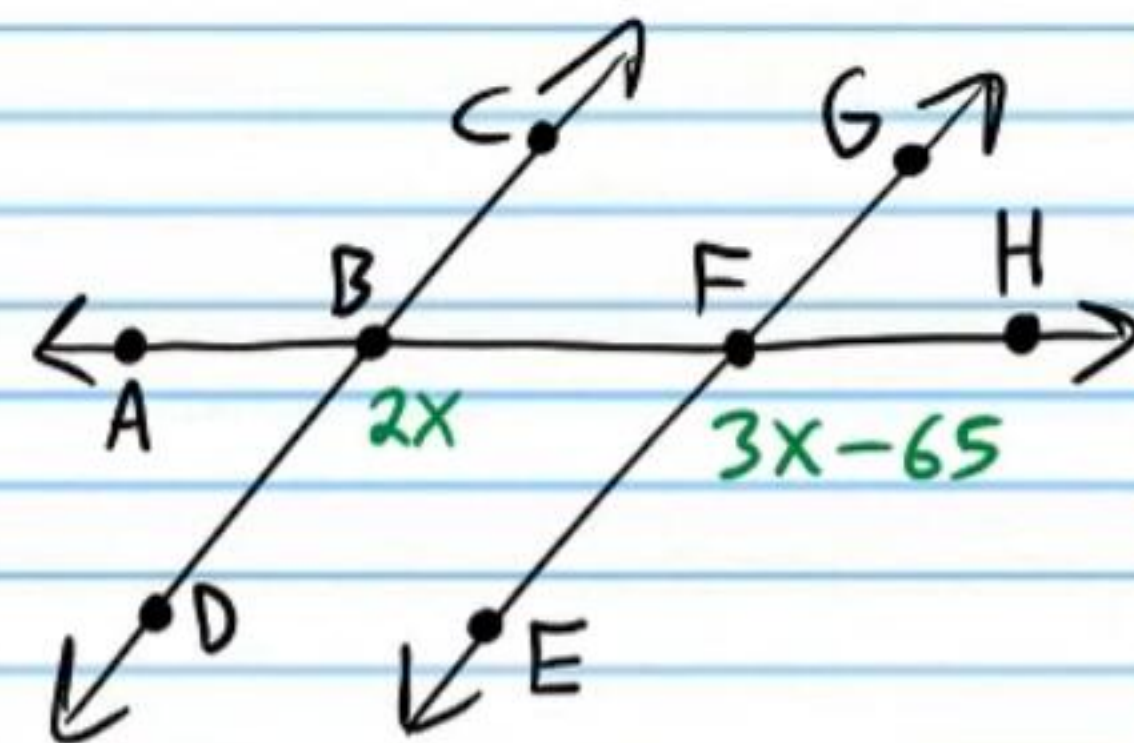
- ⑥ If $m\angle 1 = 23^\circ$ in the diagram below, then $m\angle 2 = 157^\circ$.



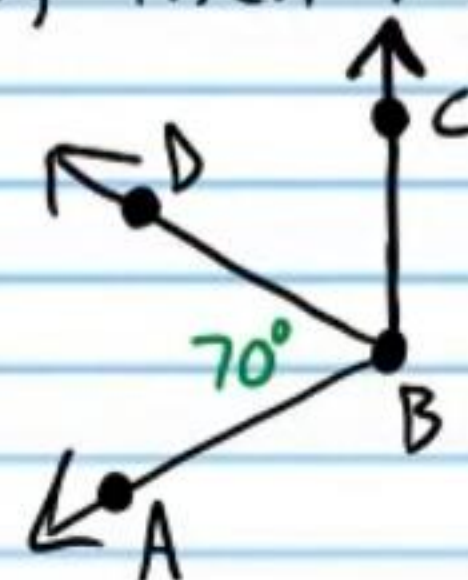
- ⑦ If $\overline{AC} \cong \overline{DE}$ in the segments below, then $DE = 48$.



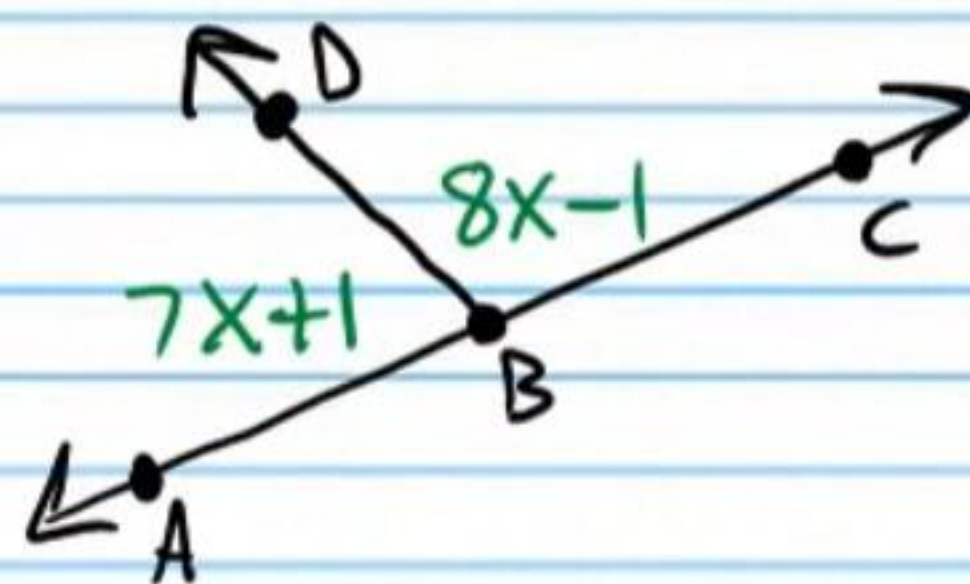
- ⑧ If $m\angle EFH = m\angle ABC$, then $x = 65$.



- ⑨ If $m\angle ABC = 120^\circ$ in the figure below, and $m\angle ABD = 70^\circ$, then $m\angle DBC = 50^\circ$.

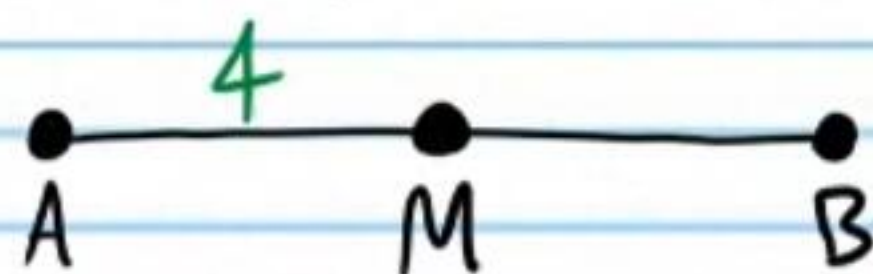


⑩ If $\angle ABD$ and $\angle DBC$ below form a linear pair, then $x = 12$.

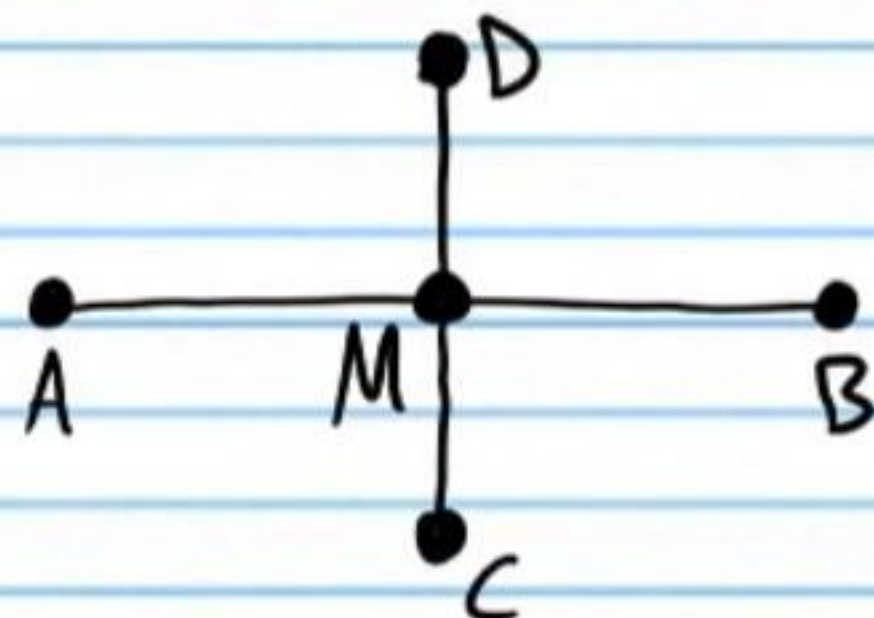


⑪ The converse of problem #6.

⑫ If M is the midpoint of the segment below, and $AM = 4$, then $AB = 8$.



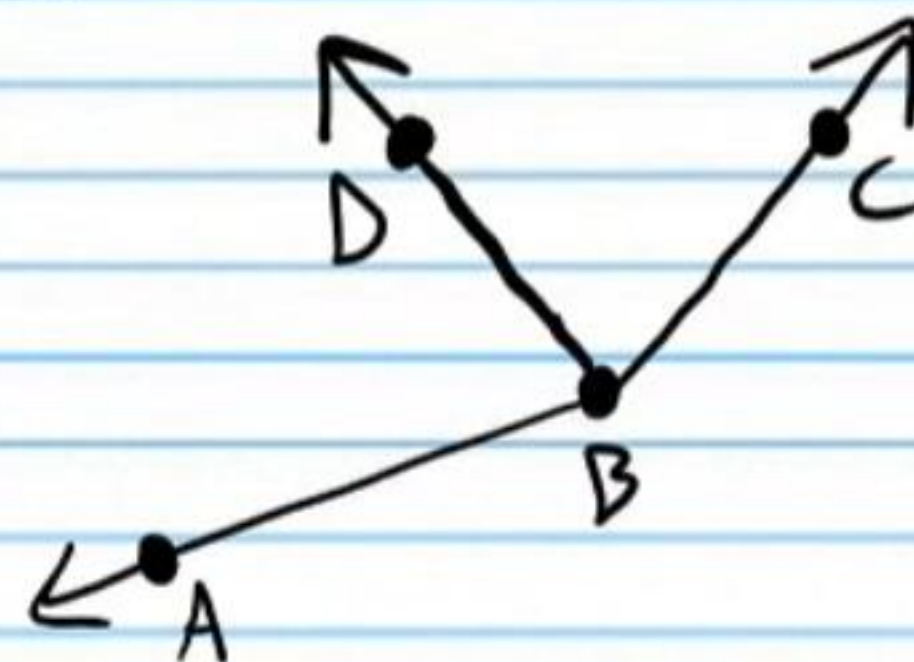
⑬ If $AB = 18$ below and $MB = 9$, then $\overline{CD}$ bisects $\overline{AB}$.



⑭ If $AB = 2MB$, then M is the midpoint of $\overline{AB}$ (this is the converse of problem #17 in class).



⑮ If $m\angle ABD = 80^\circ$ and $m\angle DBC = 80^\circ$, then $\overrightarrow{BD}$ bisects $\angle ABC$.



Answers

① Statement	Reason
I) $AB = 15$ & $BC = 3$	Given.
II) $AC = AB + BC$	Seg Add. Post.
III) $AC = 15 + 3$	Subs. Prop of =.
IV) $AC = 18$	Simplification.

② Statement	Reason
I) $\angle ABE$ & $\angle CBD$ are vert. $\angle$'s.	Given.
II) $m\angle ABE = m\angle CBD$	Vert. $\angle$'s Thm
III) $m\angle ABE = 180 - 2x$ and $m\angle CBD = 5x - 30$	Given (diagram).
IV) $180 - 2x = 5x - 30$	Subs. Prop of =.
V) $180 = 7x - 30$	Add Prop of =.
VI) $210 = 7x$	Add Prop of =.
VII) $30 = x$	Div. Prop of =.
VIII) $x = 30$	Symm. Prop of =.

③ Statement	Reason
I) $m\angle ABD = 55^\circ$ & $m\angle DBC = 65^\circ$	Given.
II) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
III) $m\angle ABC = 55 + 65$	Subs. Prop of =.
IV) $m\angle ABC = 120^\circ$	Simplification.

④ Statement	Reason
I) $\overline{AB} \cong \overline{BC}$	Given.
II) $AB = BC$	Def. of $\cong$ seg
III) $\overleftrightarrow{DE}$ bisects $\overline{AC}$.	Def. of seg bisector.

⑤ Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given.
II) $AB = CD$	Def of $\cong$ seg
III) $CD = AB$	Symm Prop of =.
IV) $\overline{CD} = \overline{AB}$	Def. of $\cong$ seg

⑥ Statement	Reason
I) $\angle 1$ & $\angle 2$ form a linear pair.	Given (diagram).
II) $\angle 1$ & $\angle 2$ are supplementary.	Linear Pair Post.
III) $m\angle 1 + m\angle 2 = 180$	Def. of supp $\angle$'s.
IV) $m\angle 1 = 23^\circ$	Given.
V) $23 + m\angle 2 = 180$	Subst. Prop of =.
VI) $m\angle 2 = 157^\circ$	Subst. Prop of =.

⑦ Statement	Reason
I) $AB = 20$ & $BC = 28$	Given (diagram).
II) $AC = AB + BC$	Seg Add Post.
III) $AC = 20 + 28$	Subst. Prop of =.
IV) $AC = 48$	Simplification.
V) $\overline{AC} \cong \overline{DE}$	Given.
VI) $AC = DE$	Def. of $\cong$ Seg
VII) $DE = 48$	Trans. Prop of =.

⑧ Statement	Reason
I) $m\angle EFH = m\angle ABC$	Given.
II) $m\angle ABC = m\angle DBF$	Vert. $\angle$'s Thm.
III) $m\angle EFH = m\angle DBF$	Trans. Prop of =.
IV) $m\angle DBF = 2x$	Given (diagram).
V) $m\angle EFH = 2x$	Trans. Prop of =.
VI) $m\angle EFH = 3x - 65$	Given (diagram).
VII) $2x = 3x - 65$	Trans. Prop of =.
VIII) $-x = -65$	Subst. Prop of =.
IX) $x = 65$	Div. Prop of =.

⑨ Statement	Reason
I) $m\angle ABC = 120^\circ$ & $m\angle ABD = 70^\circ$	Given.
II) $m\angle ABC = m\angle ABD + m\angle DBC$	$\angle$ Add Post.
III) $120 = 70 + m\angle DBC$	Subst. Prop of =.
IV) $50^\circ = m\angle DBC$	Subst. Prop of =.
V) $m\angle DBC = 50^\circ$	Symm. Prop of =.

⑩ Statement	Reason
I) $\angle ABD$ & $\angle DBC$ form a linear pair.	Given (diagram).
II) $\angle ABD$ & $\angle DBC$ are Supplementary.	Linear Pair Post.
III) $m\angle ABD + m\angle DBC = 180$	Def. of Supp $\angle$'s.
IV) $m\angle ABD = 7x+1$ and $m\angle DBC = 8x-1$.	Given (diagram).
V) $7x+1 + 8x-1 = 180$	Subs. Prop of =.
VI) $15x = 180$	Simplification.
VII) $x = 12$	Div Prop of =.

⑪ Statement	Reason
I) $\angle 1$ & $\angle 2$ form a linear pair.	Given (diagram).
II) $\angle 1$ & $\angle 2$ are Supplementary.	Linear Pair Post.
III) $m\angle 1 + m\angle 2 = 180$	Def. of Supp $\angle$'s.
IV) $m\angle 2 = 157^\circ$	Given.
V) $m\angle 1 + 157 = 180$	Subs. Prop of =.
VI) $m\angle 1 = 23^\circ$	Subt. Prop of =.

⑫ Statement	Reason
I) M is the midpoint of $\overline{AB}$.	Given.
II) $AM = MB$	Def. of midpoint.
III) $AM = 4$	Given.
IV) $MB = 4$	Trans. Prop of =.
V) $AB = AM + MB$	Seg Add Post.
VI) $AB = 4 + 4$	Subs. Prop of =.
VII) $AB = 8$	Simplification.

⑬ Statement	Reason
I) $AB = AM + MB$	Seg Add Post.
II) $AB = 18$ and $MB = 9$.	Given.
III) $18 = AM + 9$	Subs. Prop of =.
IV) $9 = AM$	Subt. Prop of =.
V) $AM = MB$	Trans. Prop of =.
VI) $\overline{CD}$ bisects $\overline{AB}$.	Def. of Seg bisector.

⑭

Statement	Reason
I) $AB = AM + MB$	Seg Add Post.
II) $AB = 2MB$	Given.
III) $AM + MB = 2MB$	Trans. Prop of =.
IV) $AM = MB$	Subt. Prop of =
V) M is the midpoint of $\overline{AB}$.	Def. of midpoint.

⑮

Statement	Reason
I) $m\angle ABD = 80^\circ$ & $m\angle DBC = 80^\circ$	Given.
II) $m\angle ABD = m\angle DBC$	Trans. Prop of =.
III) $\overrightarrow{BD}$ bisects $\angle ABC$	Def. of $\angle$ bisector.